

Foreword

The first rule of system design states: 'If you don't need to use a distributed design, then don't do so.' Distributed systems are more complex than centralised ones, take longer to design, need more code to implement and are more complex to operate. In short, they are more expensive to develop and run.

However, life is not always kind to us. Many systems cannot be designed in a centralised way for a variety of reasons. For some, the largest computers money can buy are simply not big enough to do the whole job. For others, the response time needed for some functions can only be met if the processor is very local to the user.

Increasingly, though, our world consists of a set of systems that are linked together. Even systems that were designed as large, stand-alone, centralised ones have been interconnected. The biggest of them all, CSS (customer service system), started as a stand-alone system, but soon the separate instances of it had to be linked up so that customers who dial into one office could be served from any CSS database. Connections to operational maintenance centres, for System X exchanges, and to Work Manager, for allocation of engineers for orders and faults, soon followed. Then SMART came along to provide a graphical, intelligent front end to CSS. Now CSS is also linked to systems offering voice response and Internet access to BT's customers.

Everywhere we look this picture is repeated. It is now rare to develop new systems which do not have on-line links to other systems from day one. This trend will be enhanced by the drive for e-business and e-admin. The Internet is fundamentally a distributed approach. We can no longer hope for a world where we have a few, big, separate systems. We just have to get better and better at understanding the design needs for distribution.

In such a world, computing relies on a quality operational service. We need to work hard to implement a robust infrastructure, including firewalls, authentication, directories, IP address management, if we are to have a cost-effective computing environment. Everyone needs to play a part in the end-to-end design, so we look forward to an exciting distributed future. It is vital BT excels at meeting this challenge.

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